MAKEGOOD POWER CHAIR

MakeGood Power Chair — Project Overview

What we are doing, and why

R.O.A.R. / TSSFAA with **MakeGood INC**
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Private pre-release draft. Not yet safe to build from. Prices are September 2026 US retail.

What we are doing, and why

R.O.A.R. / TSSFAA with MakeGood INC — October 2026 — pre-release, private

The problem

Children who cannot self-propel a manual chair get powered mobility late, if at all. Pediatric power chairs are expensive, insurance approval is slow, children outgrow them, and the controls assume a joystick-capable user. Meanwhile, a working hoverboard — two brushless hub motors, a battery, and a motor controller — sells used for next to nothing and gets thrown away by the thousand.

MakeGood already solved the manual half: the Toddler Mobility Trainer (TMT) is a 3D-printed, open-source mobility chair for children aged roughly one to eight, built by volunteer makers on desktop printers.

What we are building

A **powered drive add-on for the TMT** that keeps the chair usable as a manual chair, and that a volunteer maker can build from printed parts plus commodity components.

Three things make it different from "put motors on a chair":

It is dual-mode. The child self-propels on the printed wheels today. When they need power, drive wheels deploy — the same frame, the same seat, the same chair. The mechanism is a lever that lifts the printed wheels onto hub motors, or a removable power pod, or a friction drive. Power-only is not an option we offer.

Parents choose the supervision, not us. A dead-man switch is right for one family and wrong for the next. Instead of deciding, the design offers hold-to-run, e-stop remote, attendant takeover, tether kill cord, timed sessions, proximity limit, and fully independent — in any combination, saved as profiles, each with its trade-offs written down. The only fixed part is the **Safety Floor**: the chair stops on link loss, input fault, or controller fault; brakes hold on power loss; overtemperature and overvoltage are enforced. Circumstances dictate the correct solution; the design's job is to make every solution available and documented.

Assistive driving helps, and never takes over. Optional sensors and an on-board AI computer can watch for collisions, drop-offs and stairs, vehicles, tipping, and low power. They can *warn*, *slow*, or *stop with override*. They never add motion or steer. The driver's own input device always wins. We want a system that reduces injuries, not one that gets blamed for them.

How we are building it

Parts a volunteer can get

Subsystem	Primary route	Alternative
Motors	Hoverboard-class hub motors — **donated working hoverboards are a first-class source**	Standard 24 V wheelchair motors with integrated brakes
Controller	VESC or ODrive S1 — configured, never reflashed	RoboClaw for brushed wheelchair motors
Battery	Two DeWalt 20V MAX packs in series (36 V), hot-swappable in pairs	24 V wheelchair pack
Supervisor	RP2350 microcontroller running the parent profiles, inputs, and heartbeat	—
Brakes	Manual wheel locks plus solenoid pin lock or spring-applied EM brake	Motor's own EM brake (wheelchair motors)
Assist	Arduino VENTUNO Q with cameras and ranging sensors	UNO Q, or sensors-only on the supervisor
Input	Joystick, switches, single-switch scanning, Sip-N-Puff, head array	—

A donated hoverboard yields a matched pair of motors for free. We test it before teardown, inspect the motors afterward, replace its controller, and recycle its battery.

No reflashing

Every controller we recommend is configured through its own vendor tooling from a versioned config file in this repo. No custom firmware, no ST-Link, no "it works on my board." A volunteer in a different city builds the same chair.

A configurator instead of a parts list

There is no single bill of materials, because there is no single right chair. The repository ships a browser-based **configurator**: eighteen decisions (driver mass, passenger, drive mechanism, motors, controller, battery, brakes, assist, ventilator, inputs, supervision, settings interface) produce a build sheet with a parts list, cost range, warnings, and a verification checklist. Incompatible choices are greyed out before anything is bought. Every warning cites a requirement ID.

Requirements first

The configurator enforces a numbered requirements document (SHALL / SHOULD / MAY). Options were admitted only if they could be traced to a requirement. The study that produced them is written up in *Design Study Findings*; the cost picture is in *Price Matrix*.

What it costs

Seven representative builds, parts only, September 2026 US prices, computed by the configurator:

Build	Parts range
Donated hoverboard, minimum	$500 – $1,250
Daily-use 36 V: new hubs, hot-swap batteries, solenoid brake	$950 – $2,250
Wheelchair gearmotors + RoboClaw	$1,050 – $2,250
Everything on: vision assist, Sip-N-Puff, buddy seat	$1,550 – $3,400

Ranges are wide because most lines are still estimates; the anchors (motors, batteries, ODrive, RoboClaw, VENTUNO Q) are retail-listed. Printed parts, filament, and labour are excluded.

Who

MakeGood INC (makegood.design) — designers and stewards of the TMT; the pediatric mobility expertise and the maker network that builds and delivers chairs.

R.O.A.R. / TSSFAA (tssfaa.com) — Rex's Open Assistive Resources. Michael "T-Rex" Kadie: twenty-plus years of EV engineering (four world-record EVs, multiple battery management systems including an FDA-certified one for wheelchair use), open-source assistive technology (Sip-N-Puff, T-Rex Talker AAC). Drive, power, controls, and the configurator.

Where we are

Done	Open
Requirements v0.10	Regulatory position: trainer/DIY vs FDA Class II powered wheelchair
Option survey for every subsystem	Bench test VESC vs ODrive on real hoverboard hubs
Configurator with rules engine and price book	Catalog hoverboard axle/hall variants
Seven costed reference builds	Lift-lever mechanism design and force measurement
Private GitHub repository	Verify estimated prices; choose the public license

Nothing has been built yet. This is a design study, costed and cross-checked, ready for a first prototype. The repository is private while MakeGood and R.O.A.R. agree on direction and licensing; the intent is to publish under an open-source license matching the TMT's.

Why this matters

A child who can move chooses where to go. The TMT gave that to children who can push. This gives it to the ones who can't — on the same chair, at a price a volunteer group can carry, with supervision their parents chose and safety that doesn't depend on a company staying in business.

Repository: github.com/mkadie/MakeGoodPowerChair (private). Contact: trex@tssfaa.com